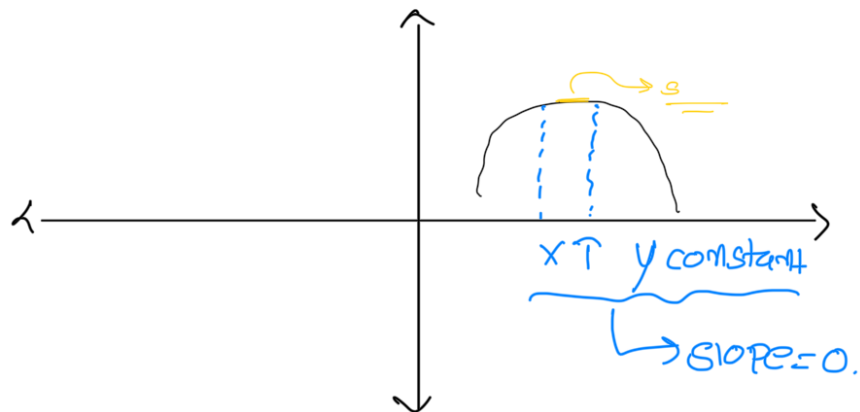
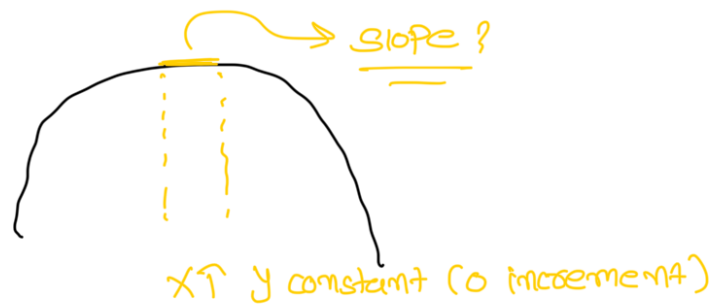
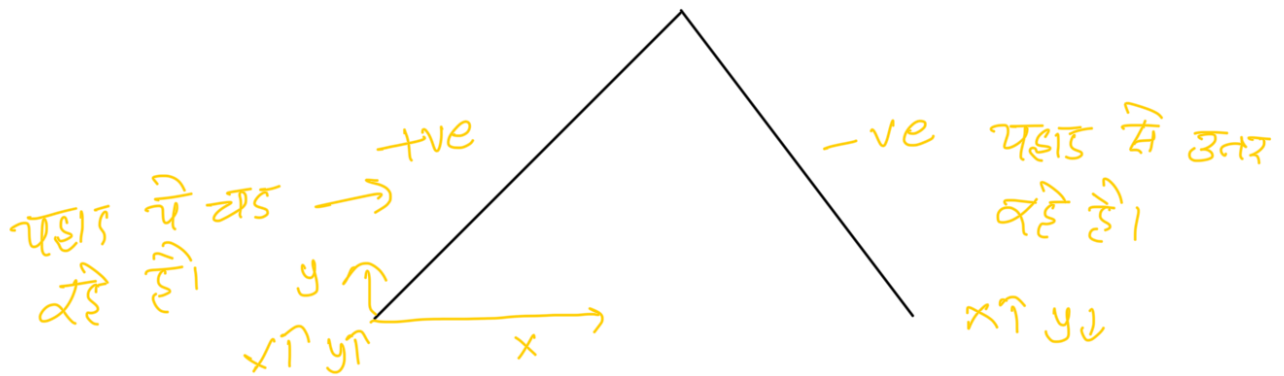


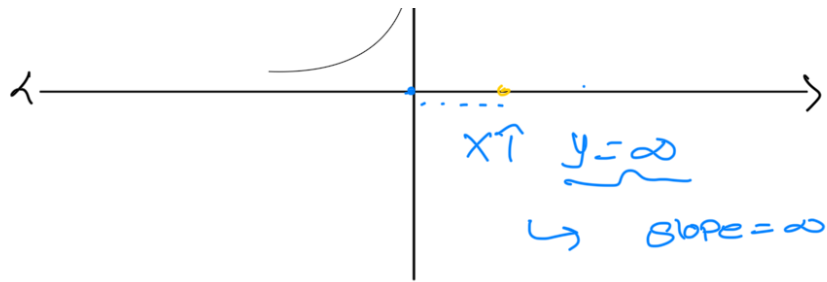
* Revision:

① Slope: Slope describes how much one variable changes in response to a change in another variable.



Slope?



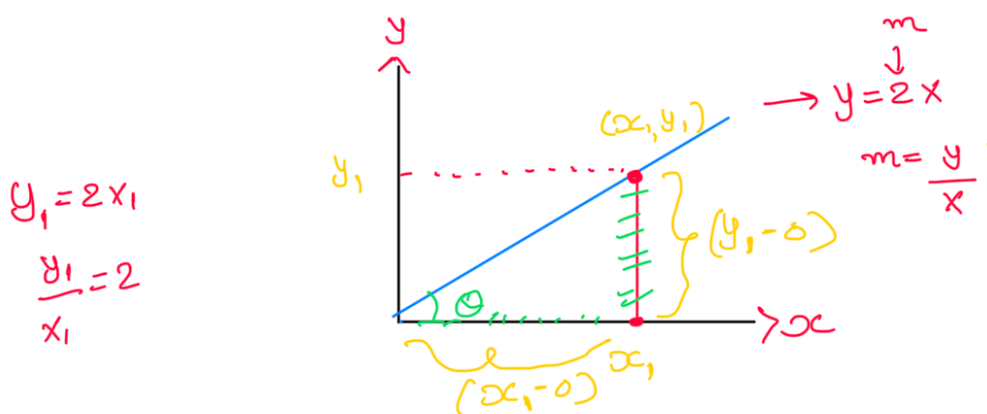


→ and how we measure slope: $\tan(\theta)$

↳ 2nd way we will see it later is

differentiation: $\frac{\partial y}{\partial x}$

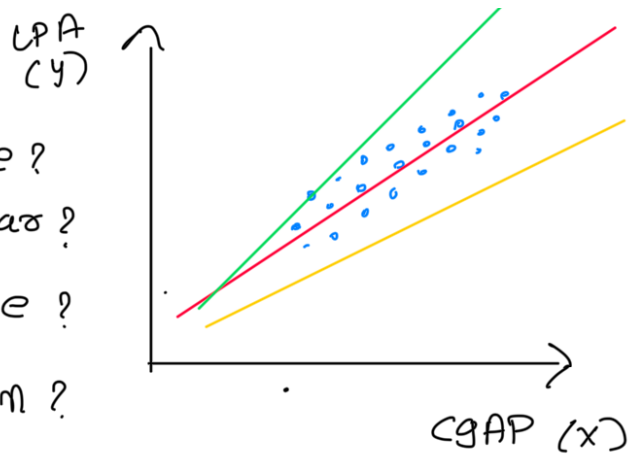
Slope describes how much one variable changes
in response to a change in another variable.



$\tan \theta = \frac{\text{opposite}}{\text{Adjacent}} = \frac{\text{change in } y}{\text{change in } x} = \frac{y_1}{x_1} = m$

② Simple Linear Regression:

Why it is simple?
 Why it is linear?
 Best prediction line?
 Line eqn equation?



- ① bcz there is only one IP feature.
- ② bcz it can draw only Line (straight)
- ③ minimum Error.
- ④ $y = mx + b$

③ Error & Regression Matrix:

a: MSE
b: RMSE
c: MAE

} Revision

d: R^2

e: Adj R^2

} we will see today.

④ Outliers: data point that is significantly different from the rest of the data.

→ effect of it: Pull the model from the actual trend.

observablehq.com

→ how to detect:
IGR
Z score
scatter plot
Box plot

} we will see this in statistics class.

→ is it useful in any cases?

↳ think about it we will answer it in end of the class.

⇒ Lecture: 5

① Multiple linear Regression:

IQ | CGPA | 12th per. | 10th per | LPA
X y
(5 features)

1 feature:

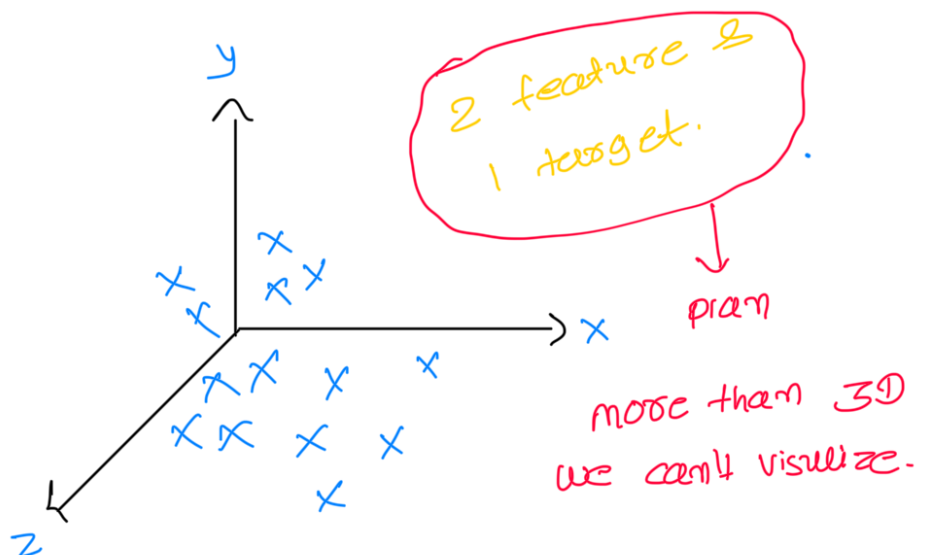
$$y = mx + b$$

2 fea:

$$y = m_1x_1 + m_2x_2 + b$$

.....

$$y = m_1x_1 + m_2x_2 + \dots + b$$



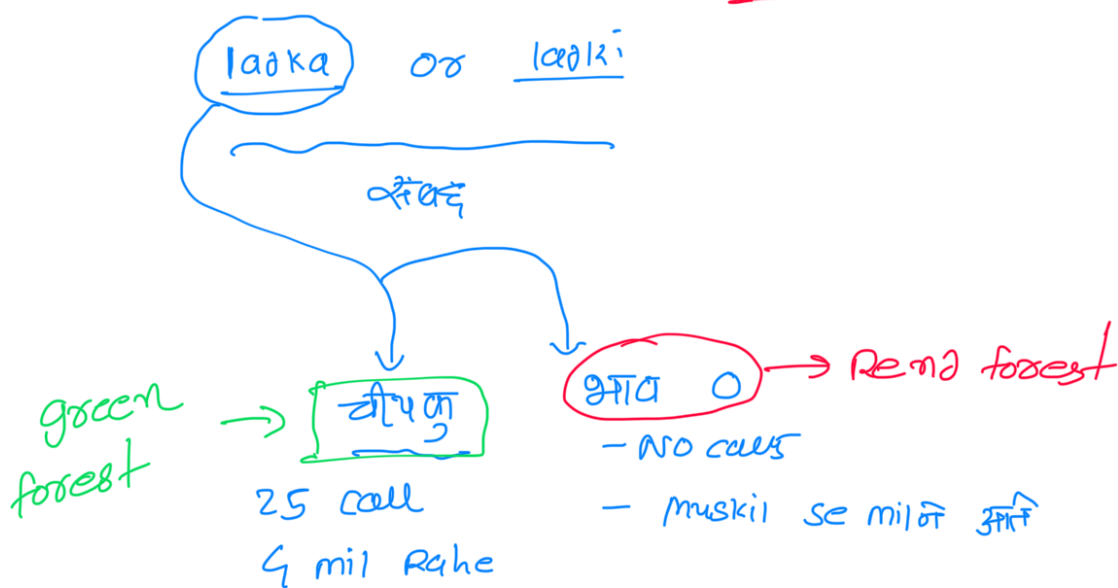
multiple-regression-
visualizer.onrender.com

Same as LR (only no of features are high.)

② Overfitting & underfitting:

over: ज्यादा

Under: कम



laaka	laaki	Relationship
<u>green forest</u>	<u>Pasam hai</u>	✓ : Perfect
Red forest	<u>Pasam hai</u>	✗ → Underfitting
green	no	✗ → overfitting
<u>Red</u>	<u>no</u>	✓ : Perfect

Rule of Relationship (healthy): $laadki$ ke jese pasand hai
Aese Rahan hain.

model should capture the behavior of the
Data.

$laadki$
↓
training
Error

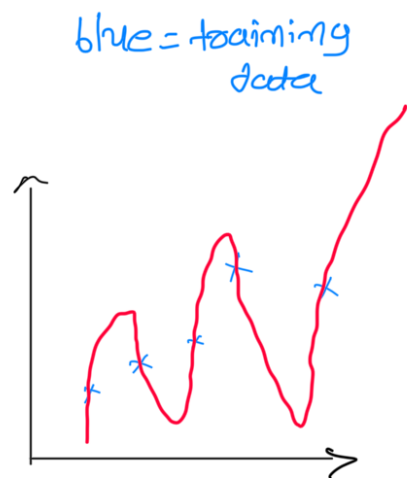
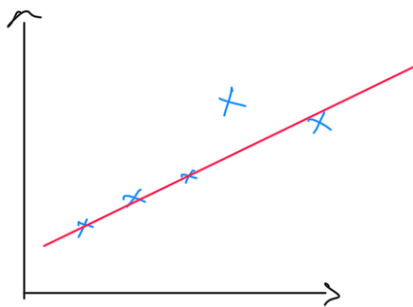
$laadki$
↓
testing
Error

training Error $\approx$ testing Error $\rightarrow$ no overfitting

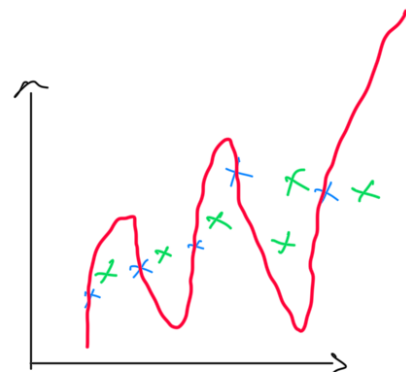
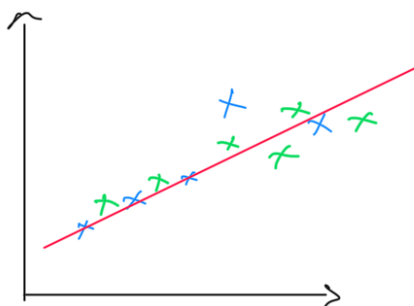
training Error $\ll$ testing Error $\rightarrow$ overfitting

training Error $\uparrow$ & testing Error $\uparrow \rightarrow$ underfitting

training Error $\downarrow$ & testing Error $\downarrow \rightarrow$ Perfect.



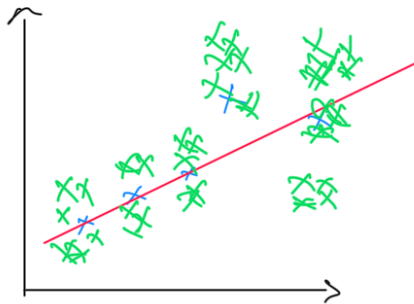
We will NOT Answer without
testing data.



↓
perfect. both low.

↑
overfitting

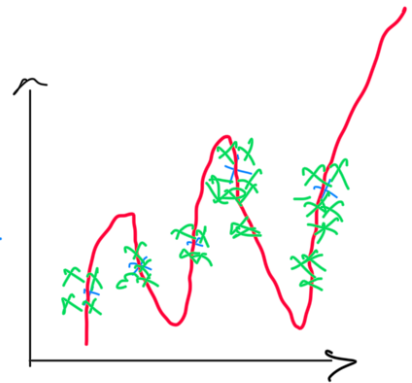
training errors low
testing " high.



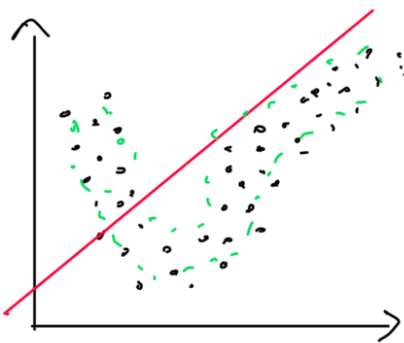
↑
overfitting

training errors ↓
testing errors ↑

→
perfect

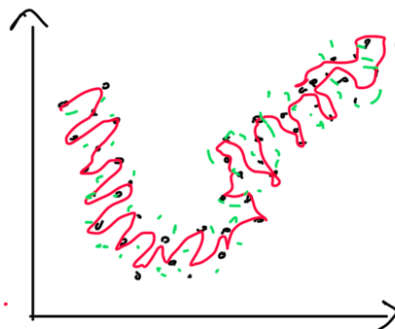


Why over/under depends on
testing data. or we can say
behaviour of data.



↑
under:

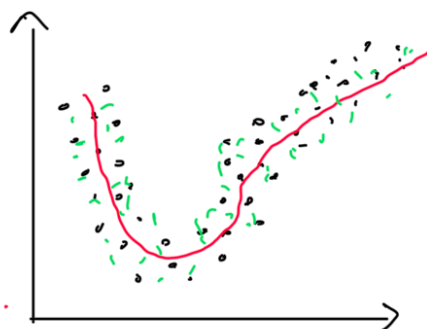
both ↑



← over:

train ↓
test ↑

green:
test
black:
training



} perfect

both ↓

Underfitting: model is too simple. (tr. ↑ test ↑)

overfitting: noisy / memorizes (tr. ↓ test ↑)

good model: perfect spot between
complex & simple. (tr. ↓ test ↓)

→ too little generalization

* When overfitting:

- for simple task : complex model (less assumption)
- Training data is too small.
- Too many features. —

→ too much generalization

* When underfitting:

- Simple model (more assumption)
- Too few features. —
- Excessive Regularization —

Q: When outliers are helpful or important ?

- Fraud detection
- Anomaly "
- Rare events